

MENTAL HEALTH & LIFESTYLE ANALYSIS

Data Analytics Project

Prepared By

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Project Objective

- Analyze mental health and lifestyle data.
- Design and build a Data Warehouse using Star Schema
- Perform the ETL process using SSIS
- Identify patterns and trends related to mental health and lifestyle
- Support data-driven decision-making through meaningful insights

Target Audience

Healthcare
Professionals

:To monitor mental health indicators, analyze stress levels, sleep patterns, and lifestyle habits, helping them better understand the factors associated with mental well-being

Researchers

:To explore relationships between demographic characteristics, lifestyle habits, and mental health conditions, supporting statistical analysis and future research

HR Departments &
Organizations

:To gain insights into work-related factors such as working hours, stress levels, and employee well-being, helping organizations create healthier work environments.

Public Health
Organizations &
Decision Makers

:To understand population-level trends, identify potential health concerns, and support awareness campaigns and public health planning.

Dataset Overview

Dataset Information

Our project is based on the Mental Health & Lifestyle Dataset, which contains information about individuals' demographic characteristics, lifestyle habits, and mental health indicators.

Main Attributes

-  Demographic Information
 - Age
 - Gender
 - Occupation
 - Country
-  Mental Health Indicators
 - Mental Health Condition
 - Severity
 - Consultation History
 - Stress Level
 - Medication Usage
-  Lifestyle Habits
 - Sleep Hours
 - Work Hours
 - Physical Activity
 - Social Media Usage
 - Diet Quality
 - Smoking Habit
 - Alcohol Consumption

Dataset Summary

- *Source: Kaggle*
- *Number of Records: 50,000*
- *Number of Features: 17*

Dataset Categories

-  ◦ Demographic Information
-  ◦ Lifestyle Habits
-  ◦ Mental Health Indicators

Data Engineering

Data Engineering Process

To prepare the dataset for analysis, we built an ETL pipeline using SQL Server Integration Services (SSIS). The processed data was then loaded into a dimensional data warehouse following the Star Schema model to .improve reporting performance and simplify analysis

ETL Process

Clean the data

Convert data type

Create derived column

Handle Slowly Changing Dimensions (SCD)

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-
-

DAX Measures

To support interactive reporting, several DAX measures were created for .KPI cards and dashboard calculations

Main Measures

Total Patients

Average Age

Average Sleep Hours

High Stress

Medication Usage

Condition Rate

-
-
-
-
-

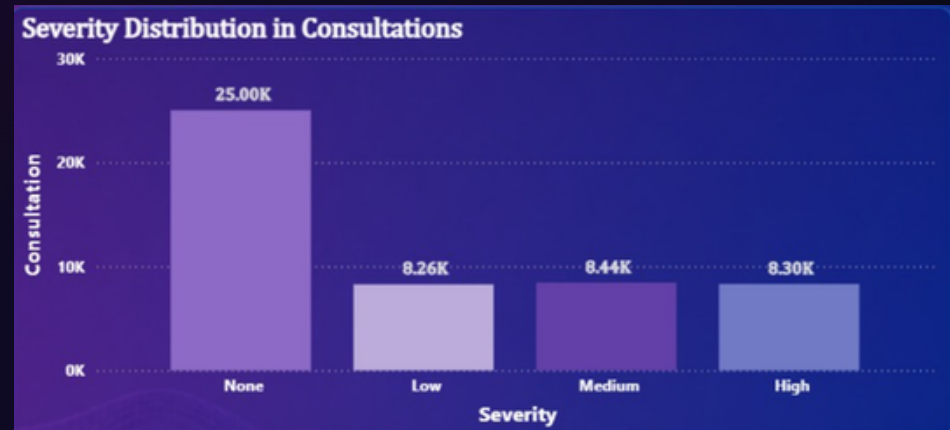
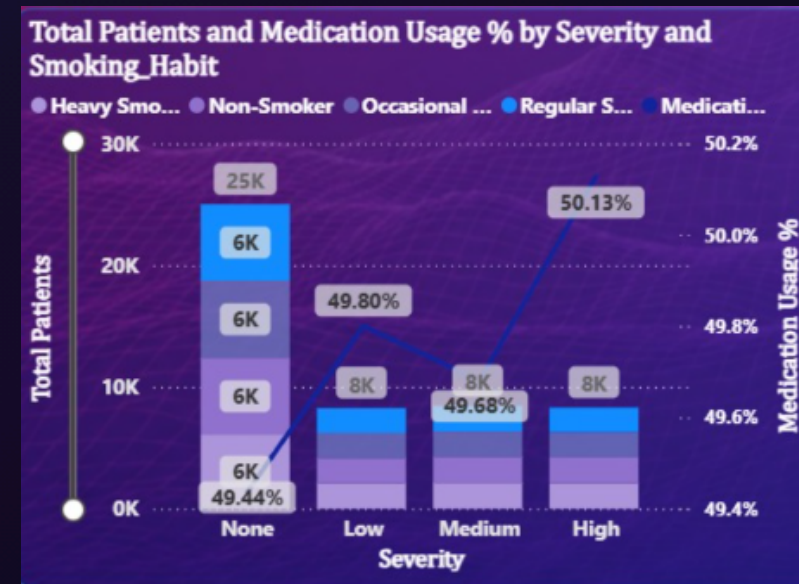


DASHBOARD OVERVIEW

The Overview Dashboard provides a high-level summary of the dataset through key performance indicators (KPIs) and interactive visualizations. It helps users quickly understand the overall distribution of respondents, demographic characteristics, and mental health indicators before exploring the data in more detail.

Key Components

- KPI Cards summarizing the most important metrics.
- Demographic distribution of respondents.
- Mental health condition overview.
- Country-wise analysis.
- Interactive filters for dynamic exploration.



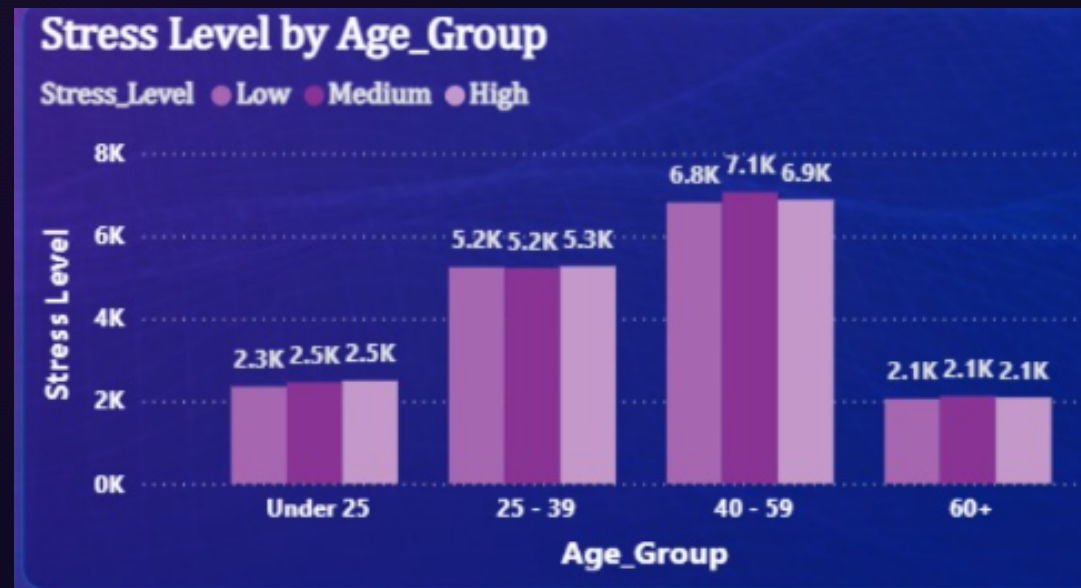
Lifestyle Analysis Dashboard

Key Insights

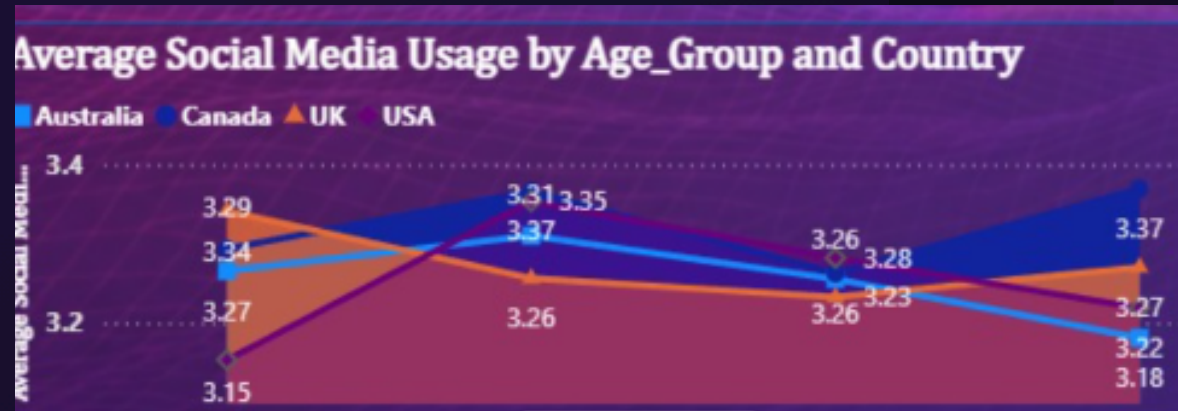
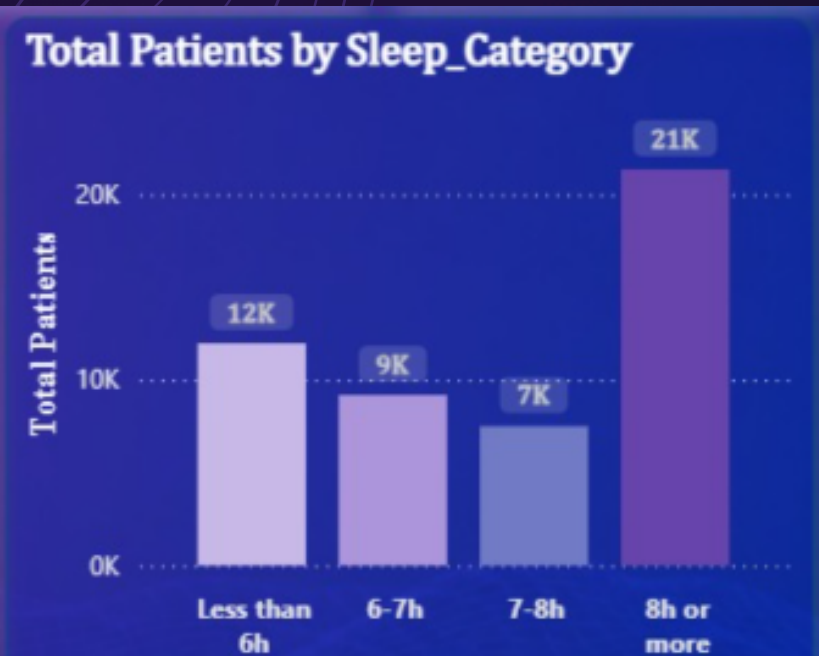
- The average sleep duration is 7.01 hours.
- 33.41% of participants experience high levels of stress.
- The risk of burnout stands at 7.85%
- Average social media usage is 3.27 hours per day.
- The majority of patients sleep 8 hours or more.
- Dietary quality is distributed relatively evenly across healthy, average, and unhealthy categories.
- Smoking rates are higher among the 40–59 age group

Purpose

This dashboard analyzes lifestyle habits and their relationship with mental health



Lifestyle Details Dashboard



Conclusion

✓ Key Insights

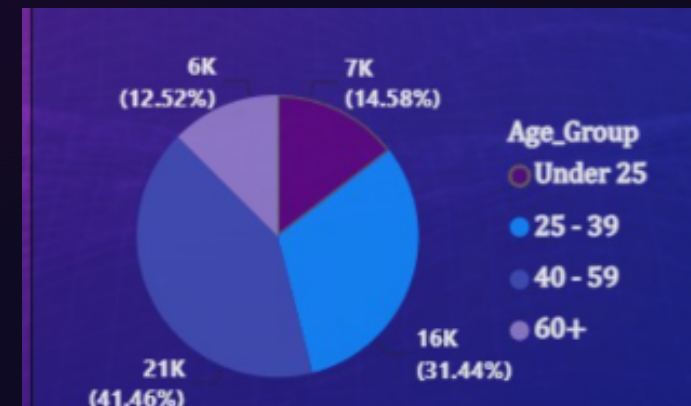
- Sleep duration is very similar across genders (around 7 hours). ✓
- Females have slightly higher average work hours than males. ✓
- Mental health conditions are distributed fairly evenly across occupations. ✓
- The 40-59 age group records the highest stress levels. ✓
- Sleep duration remains relatively stable across different age groups

Age appears to have a stronger relationship with stress than gender, while occupation shows only small differences in mental health distribution.

Health Status Dashboard

Key Insights

- Most patients fall into the "None" severity category
- Medication usage remains close to .50% across all severity levels
- Nearly 25,000 patients required mental health consultations
- Approximately half of the participants reported poor mental health conditions
- Smoking habits are represented across all severity categories



Conclusion

Through analyzing the Mental Health dataset, we found that mental health is influenced by multiple demographic, lifestyle, and health-related factors. The analysis showed that middle-aged individuals (40–59 years) experience the highest stress levels, while sleep duration remains relatively consistent across different age groups and genders. Lifestyle habits such as smoking, diet quality, and social media usage appear to be associated with stress and mental well-being. Although most participants were classified as having no severe mental health condition, consultation and medication usage remained significant, highlighting the importance of early detection and continuous mental health support. Overall, the dashboard provides clear insights that can help healthcare professionals and decision-makers better understand mental health patterns and support data-driven decisions



THANK YOU!